

## Homework 2

**35 points**

1. Let

- P: “The training dataset has more than 10,000 rows.”
- Q: “The model training takes more than 5 minutes.”

- a) Write the conditional statement  $P \rightarrow Q$ . **2 points**
- b) Is the conditional false if the model takes more than 5 minutes but the dataset has only 5,000 rows? Explain your answer. **2 points**

2. Let the original statement be “If a feature is measured in large units, then feature scaling is necessary.”  $2 + 2 + 2 = 6$

- Write the converse
- Write the inverse
- Write the contrapositive

3. Let

- P: “The data pipeline fails”
- Q: “An alert is triggered”

Which of the following is equivalent to “If the data pipeline fails then an alert is triggered”. Explain your choice. **1 + 2 = 3 points**

- a) If an alert is triggered, then the pipeline failed
- b) If the pipeline does not fail, then no alert id triggered
- c) If no alert is triggered then the pipeline did not fail

4. Prove the following using Truth Tables: **5 points**

$$(P \wedge Q) \Rightarrow R \Leftrightarrow P \Rightarrow (Q \Rightarrow R)$$

**5.** Prove the claim in question 4 using De Morgan's Law. **5 points**

6. Consider the following statement

“Removing outliers is necessary for using linear regression on this dataset”

Let

P: “Outliers are removed”

Q: “Linear regression is used”

Translate the above into a conditional statement. **2 points**

7. Prove the following using semantics: **5\*2 = 10 points**

- $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$
- $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$